

Curriculum Vitæ

Kevin Kramer

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Work and Education

- since 2023 **Postdoctoral Researcher**
Modelling of climate change impacts on Swiss agriculture
Agroscope: Federal Research Institute for Agronomy
- 2022 – 2023 **Postdoctoral Researcher**
Quantum matter and spectroscopy
Laboratory for Quantum Matter Research, University of Zurich
- 2018 – 2022 **PhD in Physics**
Quantum matter, model calculations and spectroscopy
Laboratory for Quantum Matter Research, University of Zurich
- 2016 – 2018 **Master of Science in Condensed Matter Physics**
University of Zurich with thesis at the University of Amsterdam
Thesis title: *Temperature dependent and time resolved photoluminescence of MoTe₂, MoS₂ and WSe₂*
- 2015 **Internship as Junior Scientist at CERN**
Member of the [SAFIR](#) project. Data acquisition and analysis, Monte Carlo simulations, software development in python and C++, database management, web development
- 2011 – 2014 **Bachelor of Science in Physics**
University of Zurich
Thesis title: *Measurement of the ζ -Potential in Nanofluidic Devices*

Services

- since 2021 Referee for the Journal of Open Source Software
- since 2021 Fire Brigade Obersiggenthal
- 2021 Founding member of the Institute of Physics's Sustainability Committee
- 2020 – 2022 Active member of Scientists 4 Future Switzerland
- 2018 – 2021 Junior Researcher representative in the Faculty Assembly at UZH
- Spring 2018 Civil Service: *Served on an alp. Construction of fences, leading and milking of cows, pasture maintenance.*
- 2010 Compulsory Military Service: *Served as paramedic and truck driver.*

Skills and Interests

Biogeoscience	Development and application of ecosystem models, meteorological and climatological analysis, grassland productivity
Physics	Angle-resolved photoemission spectroscopy, density functional theory, cuprate high-temperature superconductors, heavy Fermion systems
Computational methods and AI	High performance computing, simulations, machine and deep learning, parallelisation, visualization, data analysis, automation
Programming	python, R, matlab, Java, C++, Linux/UNIX, L ^A T _E X, database systems (mongodb), HTML, CSS
Experimental work	Assembly of a fluorescence microscopy setup, assembly of a cryo-photoluminescence measurement setup, 20+ ARPES beamtimes
Languages	German (native speaker), English (fluent), French (basic), Italian (basic)

Talks, Conferences and Workshops

1. Contributed talk at The European Geosciences Union's General Assembly 2024: *Integrating Sentinel-2 information into a growth model for assessing Alpine grassland dynamics under climate change*
2. Contributed talk at The International Workshop on Strong Correlations and Angle-Resolved Photoemission Spectroscopy (CORPES) 2019: *ARPES and DFT on overdoped single-layer cuprates*
3. Participation in the Materials with Novel Electronic Properties (MaNEP) Winter School 2019
4. Contributed poster at The 12th International Conference on Materials and Mechanisms of Superconductivity and High Temperature Superconductors (M²S) 2018: *Comprehensive Band Structure Study of Single-layer Cuprate Superconductors*

Publications

1. “growR: R Implementation of the Vegetation Model ModVege”. Kevin P. Kramer and Pierluigi Calanca. 2024. DOI: [10.21105/joss.06260](https://doi.org/10.21105/joss.06260). URL: <https://joss.theoj.org/papers/10.21105/joss.06260>.
2. “Revealing the Orbital Composition of Heavy Fermion Quasiparticles in CeRu₂Si₂”. Kevin P. Kramer, Rina Tazai, Karin von Arx, Masafumi Horio, Julia Küspert, Qisi Wang, et al. 2023. DOI: [10.7566/JPSJ.92.104701](https://doi.org/10.7566/JPSJ.92.104701). URL: <https://journals.jps.jp/doi/10.7566/JPSJ.92.104701>.
3. “Evidence for a Square-Square Vortex Lattice Transition in a High- T_c Cuprate Superconductor”. D. J. Campbell, M. Frachet, S. Benhabib, I. Gilmutdinov, C. Proust, T. Kurosawa, et al. 2022. DOI: [10.1103/PhysRevLett.129.067001](https://doi.org/10.1103/PhysRevLett.129.067001). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.129.067001>.
4. “Pseudogap Suppression by Competition with Superconductivity in La-based Cuprates”. J. Küspert, R. Cohn Wagner, C. Lin, K. von Arx, Q. Wang, K. Kramer, et al. 2022. DOI: [10.1103/PhysRevResearch.4.043015](https://doi.org/10.1103/PhysRevResearch.4.043015). URL: <https://link.aps.org/doi/10.1103/PhysRevResearch.4.043015>.
5. “Electronic Reconstruction Forming a C₂-symmetric Dirac Semimetal in Ca₃Ru₂O₇ - Npj Quantum Materials”. M. Horio, Q. Wang, V. Granata, K. P. Kramer, Y. Sassa, S. Jöhr, et al. 2021. DOI: [10.1038/s41535-021-00328-3](https://doi.org/10.1038/s41535-021-00328-3). URL: <https://www.nature.com/articles/s41535-021-00328-3>.
6. “Visualization of Multi-Dimensional Data – The Data-Slicer Package”. Kevin Kramer and Johan Chang. 2021. DOI: [10.21105/joss.02969](https://doi.org/10.21105/joss.02969). URL: <https://joss.theoj.org/papers/10.21105/joss.02969>.
7. “Oxide Fermi Liquid Universality Revealed by Electron Spectroscopy”. M. Horio, K. P. Kramer, Q. Wang, A. Zaidan, K. von Arx, D. Sutter, et al. 2020. DOI: [10.1103/PhysRevB.102.245153](https://doi.org/10.1103/PhysRevB.102.245153). URL: <https://link.aps.org/doi/10.1103/PhysRevB.102.245153>.
8. “Band Structure of Overdoped Cuprate Superconductors: Density Functional Theory Matching Experiments”. K. P. Kramer, M. Horio, S. S. Tsirkin, Y. Sassa, K. Hauser, C. E. Matt, et al. 2019. DOI: [10.1103/PhysRevB.99.224509](https://doi.org/10.1103/PhysRevB.99.224509). URL: <https://link.aps.org/doi/10.1103/PhysRevB.99.224509>.
9. “Three-Dimensional Fermi Surface of Overdoped La-Based Cuprates”. M. Horio, K. Hauser, Y. Sassa, Z. Mingazheva, D. Sutter, K. Kramer, et al. 2018. DOI: [10.1103/PhysRevLett.121.077004](https://doi.org/10.1103/PhysRevLett.121.077004). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.121.077004>.
10. “Two-Dimensional Type-II Dirac Fermions in Layered Oxides”. M. Horio, C. E. Matt, K. Kramer, D. Sutter, A. M. Cook, Y. Sassa, et al. 2018. DOI: [10.1038/s41467-018-05715-2](https://doi.org/10.1038/s41467-018-05715-2). URL: <https://www.nature.com/articles/s41467-018-05715-2>.

11. Robert Becker, Jean-Pierre Cachemiche, Chiara Casella, Günther Dissertori, Jannis Fischer, Alexander S. Howard, et al. “Monte-Carlo Simulation Based Estimation of NECR, Sensitivity, and Spatial Resolution of a Novel Preclinical PET Insert for MR”. In: *2015 IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC)*. Oct. 2015, pp. 1–3. DOI: [10.1109/NSSMIC.2015.7582128](https://doi.org/10.1109/NSSMIC.2015.7582128). URL: <https://ieeexplore.ieee.org/document/7582128> (visited on 05/06/2024).